

Advisor Meeting Brief — Prof. Michael Yang

Capstone: Orallexa — *From Paper to Production: Calibration Discipline in a Solo-Operated AI Trading Agent*

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One-line frame for the whole meeting: the project's pre-registered falsification gate did its job — it caught a backtest that didn't reproduce out-of-sample, so real money was correctly held. That honest result is exactly what this capstone was built to produce.

PART 1 — Proposal recap (Week 2, to re-anchor him)

Problem. Most AI-trading work reports calibration metrics (Brier, ECE) on *closed backtests* where the answer key is known. What's rarely tested is whether those calibration claims survive the *wild* — live, adversarial markets with staleness, latency, and other agents on the same signal.

Research question. Can a single-developer AI agent hold a measurable Brier-score advantage over a market-implied baseline when deployed live with small real capital (\$300–\$5k), across two heterogeneous asset classes?

Hypothesis. A multi-source pipeline (Claude probability extraction + technical indicators + Brier-feedback loop) will cut Brier by ≥ 0.05 vs a market baseline over a 90-day live window — *if and only if* live paper P&L tracks backtest for ≥ 14 days before any real money. (The "iff" is the falsification criterion.)

Methodology. - **Two test surfaces:** Polymarket binary events (baseline = market midpoint) · SpaceX-sector equities, 8 pure-play tickers (baseline = buy-and-hold basket). - **Metric:** per prediction, Brier component $(p - o)^2$; deliverable quantity = $\text{Brier_agent} - \text{Brier_baseline}$ over rolling 30-day windows. - **Public pre-registration:** every prediction timestamped on GitHub before it resolves → no cherry-picking. - **Hard risk controls:** 3-loss kill switch · \$300 cap · no leverage · ATR-scaled stop floor (2%–15%) · sector exposure $\leq 30\%$.

Outcomes are publishable regardless of sign: positive (Brier beats baseline → methods paper) · null (Brier $\approx$ baseline → honest case study) · negative (post-mortem of *which* calibration assumption broke).

PART 2 — Week 3 progress + what to say

1. Opening — where we are

EN: "Quick progress check-in. We're in Week 3. Proposal done; just finished this week's deliverable — an annotated bibliography across calibration, prediction markets, and agent self-improvement. The agent's run live the whole time, producing daily Brier-scored predictions. I have one important early result that's exactly what the methodology was built to catch."

中文: Week 3 汇报。提案完成、本周注释参考文献刚交。agent 一直 live 跑、每天 Brier 打分。有个重要早期发现想讲——正是方法论要抓的。

2. The key finding — lead with this

EN: "My proposal had an 'if and only if' clause: no real money unless live paper tracks the backtest. Two weeks in, it didn't. A walk-forward out-of-sample test came back with a negative mean OOS Sharpe — the backtest claims did **not** reproduce on held-out data. So per my own pre-registered rule, I **held** the \$300 deployment planned for late June. The gate worked. That's a falsifiable result in the project's favor, not a failure."

中文: 提案的 "iff":paper 不 track backtest 就不投真钱。两周后没 track——walk-forward 样本外 mean OOS Sharpe 为负,backtest 没复现。按自己 pre-register 的规则,我暂缓了 6 月底 \$300 实盘。闸起作用了——对项目有利的 falsifiable 结果,不是失败。

🔑 "if and only if" gate · walk-forward · OOS Sharpe negative · held real money

3. What I'm doing about it

EN: "I traced it to entry timing, not sizing. I backtested an ATR-based stop on a 30-day, $n=128$ window: baseline lost money at a 22.7% win rate; adding $1.5\times$ ATR stops swung it to 60.9% — about a \$6,800 improvement, significant at $p < 0.005$. It's in production now. I will **not** revisit real money until live paper reproduces that for ≥ 2 weeks. Measurement first, capital second."

中文: 定位到是 entry timing 不是 sizing。ATR 止损回测 $n=128$:baseline 亏、22.7% win;加 $1.5\times$ ATR → 60.9%、约 +\$6,800、 $p<0.005$ 。已上生产。live paper 不复现两周,绝不碰真钱。

🔑 $n=128 \cdot 22.7\% \rightarrow 60.9\% \cdot +\$6,800 \cdot p<0.005$

4. On schedule

EN: "On pace with the course: Week 1 interest statement, Week 2 proposal, Week 3 annotated bibliography — done. Next is the project plan and Gantt chart from my 12-week deliverable list. Every prediction is still timestamped publicly before it resolves."

中文: 进度按表:Week1-3 都完成,下周甘特图。预测仍全程公开 timestamp。

5. The ask

EN: "Two things I'd value. First, a methods gut-check: given the OOS reproduction problem, does my plan — fix entry rules, re-validate on paper, only then revisit small real capital — read as rigorous, or would you push the design elsewhere? Second, any paper on out-of-sample validation or forecast calibration you'd point me to, I'd read it this week. [If pending:] And whenever you're comfortable, I'd love to formalize your advisor signature for capstone credit."

中文: ① 方法 gut-check:修 entry → paper 重验 → 才碰小额真钱,够严谨吗?② 推 OOS validation / 校准的 paper。[未签] 方便时正式签字算学分。

Demo order

When	Pull up
§1	Annotated bibliography (this week's deliverable)
§2	Walk-forward verdict + latest weekly Brier report
§3	ATR-stops validation (n=128, +\$6.8k swing)
§4	https://alex-jb.github.io/spacex-ipo-tracker — public timestamped picks

Anticipated pushback → answer

- **"So it's not working?"** → "The *trading edge* isn't proven yet — but the *methodology* is working exactly as designed. The point was to test whether backtest calibration survives live and to not risk capital on an unproven claim. Catching that early is the contribution."
- **"Will you have a result by August?"** → "Yes, any sign. Positive → methods paper; null → honest case study; negative → documented post-mortem. All publishable; none depend on returns."
- **"Is this just wrapping GPT?"** → "No. The LLM is one signal. The agent composes LLM + technical indicators + market-implied prices into Kelly-fractional sizing via a Brier-feedback loop. The contribution is the composition + calibration layer."

Words: avoid → use

trading bot → **calibration-disciplined agent** · make money → **measure live Brier vs backtest** · it failed → **the falsification gate triggered** · side project → **production system**

Nervous-fallback one-liner

"My pre-registered gate caught a backtest that didn't reproduce out-of-sample, so I correctly held real capital. That honest negative is the kind of result this capstone was built to produce."